

Autodesk® Scaleform®

Scaleform 4

Scaleform

Scaleform 3

Scaleform 4.0

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1.03

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Autodesk® Scaleform® 4.2

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Autodesk Scaleform

Scaleform 4

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1 Scaleform 4

Scaleform v4	Autodesk		SDK	
UI SDK		ActionScript 3.0		Scaleform
Scaleform v4	AS 2.0	AS 3.0		
	AS3	SDK	Scaleform v4	
•	w/ 2x+	–	2.5D	
•	– Scaleform 4			Advance Display
•	– AS3		AS2	
•	Scaleform 4			
•	– AS3		(Code Base)	
•	/ –		AS3	
•	Flash			
•	– AS 2.0			AS 3.0
	(Runtime Exception)			
•	– AS 2.0			AS 3.0
•	–	ActionScript 3.0		
	(Method Closure)			
	Scaleform 4.2			Scaleform 4.2
	Scaleform 4.0	4.2	Scaleform 4.2	Scaleform 3.x
Scaleform	Scaleform 4.2			
Scaleform C++				(NameSpace)

2

2.1

```
namespace
namespace Scaleform
{
    class Foo
    {
        ...
    };
}
```

- 1)
Scaleform::Foo var;
- 2) 'using'
using namespace Scaleform;
Foo var;

C++

Scaleform

"Scaleform"

•

Scaleform

"using"

- Scaleform "SF" SF = Scaleform
(Header)
- SF "Render" "Sound" "Gfx"
- "Win32" "PS3"
"Scaleform::Render::D3D9"
"SF::D3D9::Render"

2.2 Scaleform

Scaleform (Public) Scaleform

Scaleform
Gfx

Scaleform 4

- Scaleform
 - o Gfx
 - AMP
 - XML
 - o Render
 - Math2D
 - D3D9
 - GL
 - PS3
 - ...
 - o Text
 - o Sound
 - o Alg
 - o UTF8Util

3

Scaleform

Scaleform

- “Scaleform”
- Src
- “Src/Kernel”

- Src
 - o GFx
 - AMP
 - AS2
 - AS3
 - ...
 - o Kernel
 - HeapMH
 - HeapPT
 - o Render
 - PS3
 - GL
 - D3D9
 - ...
 - o Sound
 - o XML

Src

cpp

(Public)

4 Includes

Scaleform 3.x	Scaleform 4	/Src	/Include
.cpp			

			'(convenience)	GfX.h
Scaleform	GfX_Kernel.h	Kernel		
Include				

include

```
#include "Kernel/SF_File.h"
#include "GfX/GfX_Player.h"
#include "GfX/GfX_Loader.h"
#include "GfX/GfX_Log.h"
```

```
#include "GfX_Kernel.h"
#include "GfX.h"
```

```
"PublicHeader      :  groupName"
```

groupName	GfX	Kernel	Render	AMP	...	none	groupName
GfX	GfX.h		groupName	Kernel			GfX_Kernel.h
	groupName	'none'					

—

Scaleform 4API

Scaleform
Scaleform 4API

Scaleform
include
Scaleform 3.x

Scaleform 4
Scaleform 4
Scaleform

Scaleform
“[Scaleform 4 \(Getting Started with Scaleform 4.2\)](#)”

Scaleform
Scaleform 4
Scaleform 4
Scaleform

Scaleform 3.x
Scaleform 4
Scaleform 4

Scaleform
“[Scaleform 4 \(Scaleform 4 Renderer Guide\)](#)”

Scaleform 4
(Include/GFxCompat.h)

Scaleform 3.x
Scaleform 4
typedefs
enums
defines

Scaleform 3.X
Scaleform 4
Scaleform 4

Apps/Demos/GFxPlayerTiny/GFxPlayerTinyD3D9Compat.cpp

GfxCompat.h
Scaleform 4
Gfx Loader
Scaleform 4
FontProvider
AS2Support
/ AS3Support

```
// Set Font Provider:  
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));  
loader.SetFontProvider(fontProvider);  
  
// Add AS2 Support:  
Ptr<ASSupport> pAS2Support = *new GFx::AS2Support();  
loader.SetAS2Support(pAS2Support);  
  
// Add AS3 Support:
```

```
Ptr<ASSupport> pASSupport = *new Gfx::AS3Support();
loader.SetAS3Support(pASSupport);
```

Scaleform 3.x Scaleform 4
GfxCompat.h

5.1

Scaleform 3.x 4 Scaleform 4

```
#include "Gfx_Kernel.h"
#include "Gfx.h"
#include "Gfx_Renderer_D3D9.h"      // or whatever platform you need
```

(Gfx.h) AS3 AS3 AS3
"Gfx/AS3/AS3_Global.h"
Scaleform LITE

AS3_Global.h and Obj\AS3_Obj_Global.xxx

AS3_Global.h Obj\AS3_Obj_global.xxx
AS3_Obj_Global.xxx " " ActionScript 3 swf
" " (script) GlobalObjectCPP C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++
ClassRegistrationTable
AS3_Global.h

ClassRegistrationTable include AS3_Global.h
ClassRegistrationTable

AS3 VM

5.2

GFC

SF_

Scaleform

GFX_

Scaleform 3.X	Scaleform 4
GFC_FX_VERSION_STRING	GFX_VERSION_STRING
GUNUSED	SF_UNUSED
GFC_64BIT_POINTERS	SF_64BIT_POINTERS
...	...

5.3 Namespace

Scaleform 4
'using namespace'

Scaleform::GFx::Event
using

cpp

```
namespace SF = Scaleform;
using namespace Scaleform;
using namespace Render;
using namespace GFx;
```

5.4

Scaleform 4

Scaleform 3.X	Scaleform 4
UInt	Unsigned
SInt	Int
Float	Float
...	

Scaleform 4

Scaleform 3.x

Scaleform

GFxEvent

Scaleform 4

Scaleform 4

GFx::Event

Scaleform 3.x

Scaleform 3.X	Scaleform 4	Scaleform 4
		'using namespace ...'

GFxSystem	Scaleform::GFx::System	System
GPtr	Scaleform::Ptr	Ptr
GFxMovieDef	Scaleform::GFx::MovieDef	MovieDef
GFxMovieView	Scaleform::GFx::Movie	Movie
GRendererD3D9	Scaleform::Render::D3D9::Renderer	Render::D3D9::Renderer
GFxRenderConfig	Scaleform::GFx::RenderConfig	RenderConfig
GString	Scaleform::String	String
GFxFSCCommandHandler	Scaleform::GFx::FSCCommandHandler	FSCCommandHandler
GFxLog	Scaleform::GFx::Log	Log
GFxImageCreator	Scaleform::GFx::ImageCreator	ImageCreator
GFxKey	Scaleform::GFx::Key	Key
GFxKeyEvent	Scaleform::GFx::KeyEvent	KeyEvent
GFxCharEvent	Scaleform::GFx::CharEvent	CharEvent
GFxEvent	Scaleform::GFx::Event	Event
GMatrix2D	Scaleform::Render::Matrix2x4	Matrix2x4
GMatrix3D	Scaleform::Render::Matrix3x4	Matrix3x4
GMatrix3D	Scaleform::Render::Matrix4x4	Matrix4x4
GRect		

6 Scaleform 4.0 4.2

Scaleform 4.2 4.0 API Scaleform 4.2
API 4.0

6.1

SF 4.2 PCRE Perl Perl Compatible
Regular Expressions PCRE AS3 3rdParty
Scaleform 4.2 PCRE AS3
PCRE *Include/GFxConfig.h* SF_ENABLE_PCIRE
PCRE

6.2

Scaleform 4.2 Scaleform Scaleform
(Build System)
Render Sound SF 4.2
Render Scaleform
X360/D3D9/D3D1x InitHAL()
• HALConfig_DynamicShaderCompile
Visual Studio
ShaderData.xml Render
Custom Build Step

6.3

Scaleform 4.2 -- Scaleform 4.2
40 SF 4.0
sf_license.txt Scaleform 4.0 gfxlicense.txt
gfx.lib
sf_license.txt Scaleform AMP Scaleform Exporter
Scaleform Video IME sf_video_license.txt
sf_ime_license.txt

Scaleform